Google Summer of Code project proposal - Zulip

Cluster of priority features
Focused around topics and notifications (messages) areas

Personal Information

Name	Prakhar Pratyush
Email	prakhar841301@gmail.com
GitHub Username	prakhar1144
Zulip Username (CZO)	Prakhar Pratyush
Contact Number	[redacted]
LinkedIn Username	prakhar1144
Personal Website	prakhar1144.github.io
Country of Residence	India
Timezone	Indian Standard Time (GMT+5:30)
University	National Institute of Technology Hamirpur

Abstract

This project proposal aims towards implementing high-priority features in zulip/zulip focused around topics and notifications (messages) areas. It contains a brief introduction, prior-work or discussions done, and a proposed approach for each of the mentioned features.

Previous Contributions

Code Contributions

- Pull Requests (Merged/Closed)
- Pull Requests (Open)
- Commits

1. Unmute Topics

    - #backend > Unmute Topics
    - #api design > duplicate request to update UserTopic row

    - #24236 - Merged
    - user_topic: Refactor 'do_set_user_topic_visibility_policy'. #24629 - Merged
    - user_topic: Don't raise errors on duplicate requests. #24690 - Merged
    - user_topic: Update UserTopic records regardless of the visibility_policy. #24742 - Open

2. Automate "Include organization name in subject of message notification emails" settings.

    - #api design > automate - include realm name in missedmessage email subject
    - #development help > openapi

    - user_settings: Automate 'Include realm name in missedmessage email subject' #24075 - Merged

3. Do not send email notifications for messages already present when the account is created.

    - #issues > email reply to Welcome Bot

    - email_notifications: Fix missed message email notifications of the welcome bot. #23225 - Merged

4. Send resolve topic notifications when moving streams and also resolving/unresolving a topic in a single API request

    - message_actions: Fix move messages/topic related notification-bugs. #23886 - Merged

      While working on the issue, I **discovered other bugs,** which I resolved in separate commits:
        - 8d7c15b
        - A8bcc32
        - Ca60091

5. Other Pull Requests for Bug fixes:

    - templates: Update text on 500 error page. #23120 - Merged

- test_classes: Improve assert_database_query_count output message. #23326 - Merged
- docs: Improve move content to another stream/topic instructions. #23710 - Merged
- realm_redirect: Redirect always to the login page with the next parameter. #23662 - Open
- send_email: Improve configurability for Zulip outgoing email sender name. #23917 - Open
- zilencer: Update tuple handling pattern; returned by a delete() query. #24737 - Open

Issues Reported

S.No.	Title	Status
1	#23674 - Missing input field for 'custom time' in the "delay before sending message notification emails" setting.	Fixed by #23675
2	#23325 - Improve assert_database_query_count output message.	Fixed by #23326
3	#24485 - Update 'Documenting REST API Endpoints" docs.	Fixed by #24667
4	CZO - Update "move a topic to another stream" docs to include permission details.	Fixed by #23710
5	CZO - Improve Organization settings > Bots page	Open - #24154
6	#228 - Fix zulipbot unassign operation	Open

Proposed Work

1. Follow Topics - #6027

> This is one of the five most requested features for the Zulip project overall.
> The actual product design is still to be decided, but as commented:
>
> This topic will behave like `UNMUTED`, plus some additional display and/or notifications priority that is TBD and likely to be configurable; see #6027.

After reading #6027, I put together a summary of user feedback for a "followed" topic:

a. Global notifications settings for “followed topics”.
b. Per stream notifications setting on how “followed” topics in that stream notify.
c. A new row in the notifications setting window for each “followed” topic.
d. Automatically follow a topic when you send a message. (a user setting to control this behaviour)
e. A view, or pseudo-stream is provided that contains only your “followed” topics.
f. A new keybind is provided to “go to next unread followed topic”

Approach:

a. Discussion around the exact featureset.

b. We'll make the appropriate changes in the new view (that will be made soon as part of the UNMUTE topics project), do_set_user_topic_visibility_policy, and set_user_topic_visibility_policy_in_database to make sure that setting or unsetting the FOLLOWED visibility policy works.

   We made these methods so that they don't need to be changed again, but if we find out during the product discussion that we need to add some checks or something else, we can.

c. There has been prior work by @ryanreh99 in #14555 with a TODO comment:

   > TODO: We will likely replace this query with a separate FollowedTopics table, for greater efficiency.
   > For now we assume users follow all topics to which they have posted in, but after the migration we would want to allow users to control which topics they want to follow or unfollow.

   After the introduction of visibility_policy in UserTopic (previously MutedTopic), we can allow users to control which topics they want to follow or not.

   We can use a lot of the work from the PR to implement the notification settings.

d. Most of the time, it would require modifications to do_send_messages, get_recipient_info, and user_allows_notifications_in_StreamTopic.

e. Discussion and implementation of how “followed topics” are displayed.

2. Pinned Topics - #5200, #19483

Alternative chat products have the very useful feature of pinning messages within a stream or channel. Currently, Zulip lacks this feature. Although there is an option to display stream descriptions at the top of stream narrows, it doesn’t cover all the use cases.

After discussion, Zulip is moving towards a model that allows pinning one or more topics within a stream, and then users (with permissions) can move messages into the pinned topic as needed.

Prior Work:

a. We already have an implementation plan ready after discussion in #feedback > pin topics/messages
b. We have a draft PR #21226 which covers the first two bullet points of the implementation plan.

Approach:

a. I will try to rebase the PR #21226 and polish it to make it mergeable.
b. Work on the remaining bullet points and follow-ups.

3. @here mentions - #3750, #2183

The discussion around "what wildcard @-mentions we want to support" resulted in the start of the development of the ***@here mention*** feature - #2183
It just sends desktop notifications to online users who have subscribed.

Prior Work:

a. There is a PR #16964 by @akshatdalton with substantial work on this.
b. It was blocked due to various follow-up PRs #18426, #18529

Approach:

a. A discussion around remaining blockers (if any, developed over time)
b. Once the blockers are resolved. I will try to rebase #16964 and polish it to make it mergeable.

4. @topic mentions - #3750, #22829

The discussion around "what wildcard @-mentions we want to support" and user feedbacks resulted in the start of the development of the ***@topic mention*** feature - #22829

This project adds a new wildcard that mentions the topic participants
(Anyone who sent a message to a topic or reacted to a message on the topic.)

Prior Work:

a. Discussion in #feedback > @participants about whether or not to trigger notifications in cases of move-message and message-edit (to add @topic)

operation.

b. An inactive PR #22855

Approach:

a. I will try to rebase the inactive PR #22855 and address the review comments. (It might make sense to close the PR after discussion as the review comments suggest a different approach altogether.)

b. Broadly, at the end of this feature we should have:
   - A @topic wildcard with the intended behaviour
   - a notification setting to control @topic mention.
   - @-mention typeahead for @topic

5. Email/Push notifications for alert words - #13127, #5137

Currently, email/push notifications are not sent for alert words, only desktop notifications. We need to make this possible and offer user settings to control the behaviour.

Prior Work:

a. There is an inactive PR #13458 with an attempt to implement email notifications for alert words.

Approach:

a. We would need to make most of the changes around `UserMessageNotificationsData` to add `alert_word_email_notify`, `alert_word_push_notify`, and `NotificationTriggers.ALERT_WORD`.

b. `flags` contains the data regarding `has_alert_word`, and `send_message_request.rendering_result.user_ids_with_alert_words` provides `user_ids_with_alert_words`.

c. My previous experience (#24075) implementing a user_settings will help.

6. Other priority features:

These are some of the other interesting, high-priority features. **I would try to work on these during GSoC (while waiting for reviews), but if I don't have enough time, I will pick them up after GSoC.**

a. Add an option for whether to receive a PM when invited users join. - #20398
b. Blank notification is triggered when someone sends an image. - #8087, #8796
c. Add end-to-end encryption for push notifications. - #6954

Availability

I have my university exams scheduled (May 4–13), during which I will mostly be involved in product and technical discussions in CZO. After that, I have no other commitments for the contribution period, so I would like to take on the **large 350-hour project**. I plan to work an average of 30-35 hours per week for the 12-week duration.

Relevant Experiences

1. I have about 3 years of experience working with HTML, CSS, JS, Django, and full-stack web development as a whole.

2. Previously, I have worked as a backend developer intern at a Mumbai, India-based Ed-Tech startup, Ques10.

   Django, the Django REST Framework, HTML, Bootstrap, JS, and MySQL comprised the technology stack.

   This readme briefly documents my day-to-day learnings.

3. I won the award for Best Overall Design at a HINT 5.0 hackathon with a team of two people. My primary responsibility was the design and integration of the frontend with the smart contract. - website link

   React, Material UI, Solidity, and other minor JS libraries comprised the technology architecture.

4. I also love to answer questions on StackOverflow in my free time and write some blogs and tutorials. (The technologies are around the tech stack that Zuilp uses.)
   a. StackOverflow: ~91k people reached
   b. GeeksforGeeks: ~20k views

5. As a member of my college's Google Developer Student Club's web team, I helped run events, workshops, and projects that had to do with web technologies.

   The prior experience will help me a lot during GSoC to communicate my thought process to others.

6. Personal Project: CarPool · React, Material UI, Django REST Framework, Django Channels

Why Zulip

This message by Tim Abbott inspired me a lot to stay connected to an organisation like Zulip. especially the following points:

- I want to prove that one can build a great business that is entirely about making the world's best software, all open source.
- When humanity goes to space, I want us to bring the best software with us, and proprietary cloud is not going to be portable.

Since I was introduced to the concept of open source as a freshman, I have adored it. I participated in university-level open source events and attended talks and workshops.

In October of last year, however, I eventually had the leisure to investigate open source in greater depth.

Initially, I decided to contribute to Zulip because I've always admired the engineering behind chat products, and a portion of the tech stack matched what I already knew. (Previously, I had made a personal project involving a chat feature using WebSockets.)

The detailed documentation, public discussion around product design and technical design, and a very helpful community of great engineers and contributors all add up to a great reason to keep contributing to Zulip.

I have learned the importance of writing clean commit messages, unit testing concepts, a rebase-oriented git workflow, comments in codebases, sharing my thoughts in an efficient way with fellow developers, iterating on pull requests (PRs) after code reviews, and a lot more. excited for new learnings.

I aspire to be a long-term, valuable contributor to Zulip.

Timeline (Approximate)

May 4 - 13	- University Exams - I will start product and technical discussion around the "Follow Topics" feature.
May 14 - 28	- Do the preparatory changes (if required) to the code written during the UNMUTE topic project. - Start working on the notification aspect of the “Follow Topics”

	feature. - In parallel, start discussion around any blockers for the “@here mention” feature. - Rebase #16964 and polish it to make it mergeable. I will work on this while awaiting some review or discussion around the “Follow Topic” feature.
May 29 - June 11	- Complete the work on the notification aspect of the “Follow Topics” feature and discuss the plan for the display aspect. - Start implementation of the display aspect of the “Follow Topic” feature. - Work on follow-up tasks (most likely to occur) to #16964 - Start discussion around the “@topic mention” feature.
June 12 - 26	- Work on the remaining follow-ups for the “Follow Topic” feature. - Shift the primary focus onto the “@topic mention” feature - Prepare a technical plan for the “@topic mention” feature. - Implement the @topic wildcard with the intended behaviour.
June 27 - July 10	- Implement the notification setting to control @topic mention. - Work on @-mention typeahead for @topic - In parallel, I will work on implementing “email/push notifications for alert words”, which I think would be quick to finish.
July 10 - 14	Mid Term Evaluation
July 15 - 29	- Address feedback from the mentor for improvements. - I will start my work on the "Pinned Topic” feature, for which the implementation plan is already prepared. - For the “Pinned Topic” feature, I will focus on rebasing and polishing #21226 to make it mergeable. - Work on tweaking populate_db and adding UI code to sort

	pinned topics to the top.
July 29 - August 11	- Add a new organisation permission named pinned_topics_policy that controls who can pin topics within the organisation. - Start the API design discussion around the API endpoint for the “Pinned Topic” feature. - Add the new API endpoint for the "Pinned Topic" feature. - Add a new menu item to the left sidebar three-dot menu for a topic named "Pin Topic".
August 12 - 20	- Use this time to complete any follow-up or pending work. - There is a high probability that the last bit of work remaining for the “Pinned Topic” feature will be completed soon. I will primarily focus on completing this.
August 21 - 28	Final Week - Prepare and submit the final work product.

In addition to the tasks listed, I will often review pull requests that other community members have written. I’m always open to working on some other priority features that come up during the summer.